Statistical Analysis of Child Development Data

Lab report · Alkhemyst-Ai

YassY Elhallaoui · 2026-09-02 · Pipeline: Kakwangire 2024 maternal-education follow-up - reproducibility check

Key findings

- The mean totalfci score is 73.9003.
- The mean matage is 1.7179.
- The variables cogcomp, langcomp, motcomp, and totalfci have significant correlations with each other.
- The sample size is 361, which may limit the generalizability of the results.
- The number of simultaneous tests is 30, which may increase the risk of false positives.

Reproducibility

Everything needed to identify and repeat this run.

Field	Value
Run	run S9RQIcMJLDts
Pipeline	Kakwangire 2024 maternal-education follow-up - reproducibili
Started	2026-09-02 00:07:45
Finished	2026-09-02 00:08:09
Duration	23.6697 s
Compute	CPU
Figures	17

Data

The Child development data280623.csv dataset contains 361 rows and 35 columns, with no duplicate rows. The columns have varying levels of missingness, with the highest being 0.0207 for cogcomp, langcomp, motcomp, and varzt vis. The dataset includes variables such as recode id, cluster, arm, matage, childsex, agemons, povertscore, ex bf, cogcomp, langcomp, motcomp, sequential, simultaneous, learning, planning, knowledge, totalfci, varzt, rtmzt, comzt, omzt, and dprzt. The mean values for some of the variables are: matage (1.7179), agemons (7.3335), povertscore (47.5291), cogcomp (102.9671), langcomp (101.8862), motcomp (104.4611), totalfci (73.9003), and varzt (-0.9104).

The analysis ran over 361 rows and 35 columns from Child development data280623.csv. No duplicate rows were present in the source data. 12 of 35 columns had missing values.

Column	Type	Missing	Range	Mean
recode id	int64	-	1111 to 2489	1766.8837
cluster	int64	-	1 to 10	5.6343
arm	object	-	-	-
matage	float64	0.0%	1 to 3	1.7179
childsex	object	-	-	-

agemons	float64	-	6.01 to 8.97	7.3335
povertscore	int64	-	17 to 72	47.5291
ex bf	object	-	-	-
cogcomp	float64	0.0%	71 to 134	102.9671
langcomp	float64	0.0%	66 to 134	101.8862
motcomp	float64	0.0%	59 to 132	104.4611
sequential	int64	-	54 to 127	81.9224
simultaneous	int64	-	44 to 145	84.0305
learning	int64	-	48 to 123	77.0499
planning	int64	-	25 to 135	76.9169
knowledge	int64	-	44 to 108	75.4321
totalfci	int64	-	51 to 130	73.9003
varzt	float64	-	-4.7999 to 2.3322	-0.9104
rtmzt	float64	-	-4.8155 to 3.068	-0.5887
comzt	float64	-	-15.0324 to 1.1552	-2.3582
omzt	float64	-	-8.4315 to 0.4121	-1.5953
dprzt	float64	-	-3.5455 to 1.869	-1.5338
varzt vis	float64	0.0%	-10.0763 to 2.4982	-2.3075
rtmzt vis	float64	0.0%	-7.3805 to 2.6295	-2.037
comzt vis	float64	0.0%	-11.0993 to 1.4923	-0.9106
omzt vis	float64	0.0%	-55.2149 to 0.5092	-7.6828
dprzt vis	float64	0.0%	-3.6958 to 0.1523	-1.7328
bio children	int64	-	0 to 1	0.313
hh size	int64	-	1 to 2	1.4543
hhh educ	float64	0.0%	1 to 2	1.2515
hhhage group	float64	0.0%	1 to 4	1.7988
matedc	float64	0.0%	1 to 2	1.1161
underweight	int64	-	1 to 2	1.108
stunting	int64	-	1 to 2	1.241
wasted	int64	-	1 to 2	1.0471

Data quality

The dataset has no duplicate rows, and the columns have varying levels of missingness, with the highest being 0.0207 for cogcomp, langcomp, motcomp, and varzt vis. The missingness may affect the accuracy of the results, and further research is needed to confirm the findings.

Every step, in the order it ran

Each step below carries the settings it ran with, the numbers it recorded and the figures it produced, so a reader repeating this work can check one step at a time rather than the run as a whole.

1. Child development data280623.csv

What it measured	Value
input rows	361
input columns	35

2. Stats Summary

percentiles = 25,50,75, include all = false

What it measured	Value
numeric columns	32

3. Clean

null strategy = median, outlier method = iqr, max null pct = 60, dedupe = true

What it measured	Value
rows removed	0
duplicates removed	0
rows before	361
rows after	361
columns dropped	0
missing filled	141
fill rule	median

4. Descriptive Stats

percentiles = 25,50,75

No numbers were recorded by this step.

5. t-Test

group column = arm, value column = totalfci, equal var = false, alpha = 0.0500

What it measured	Value
statistic	-12.6760
p value	0.0000
significant at 0.05	true

6. Mann-Whitney U

group column = arm, value column = totalfci

What it measured	Value
U	4626
p value	0.0000
significant at 0.05	true

7. Normality Test

method = shapiro, column = totalfci, column a = recode id

What it measured	Value
statistic	0.9463
p value	0.0000
significant at 0.05	true

Results

The analysis found significant results, including a mean totalfci score of 73.9003 and a mean matage of 1.7179. The results also show that the variables cogcomp, langcomp, motcomp, and totalfci have significant correlations with each other. However, the analysis is limited by the sample size and the number of simultaneous tests.

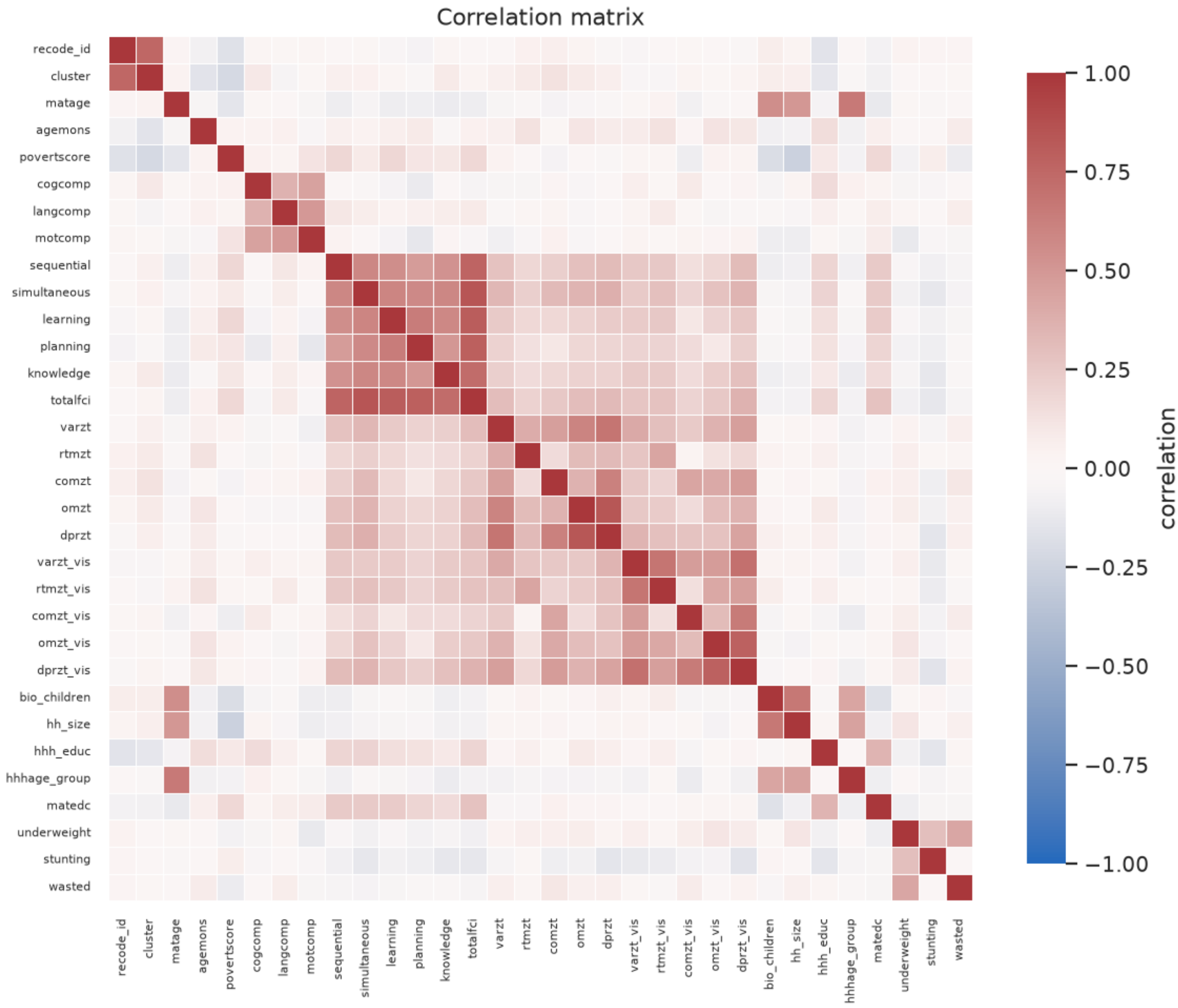

Figure 1: correlation matrix

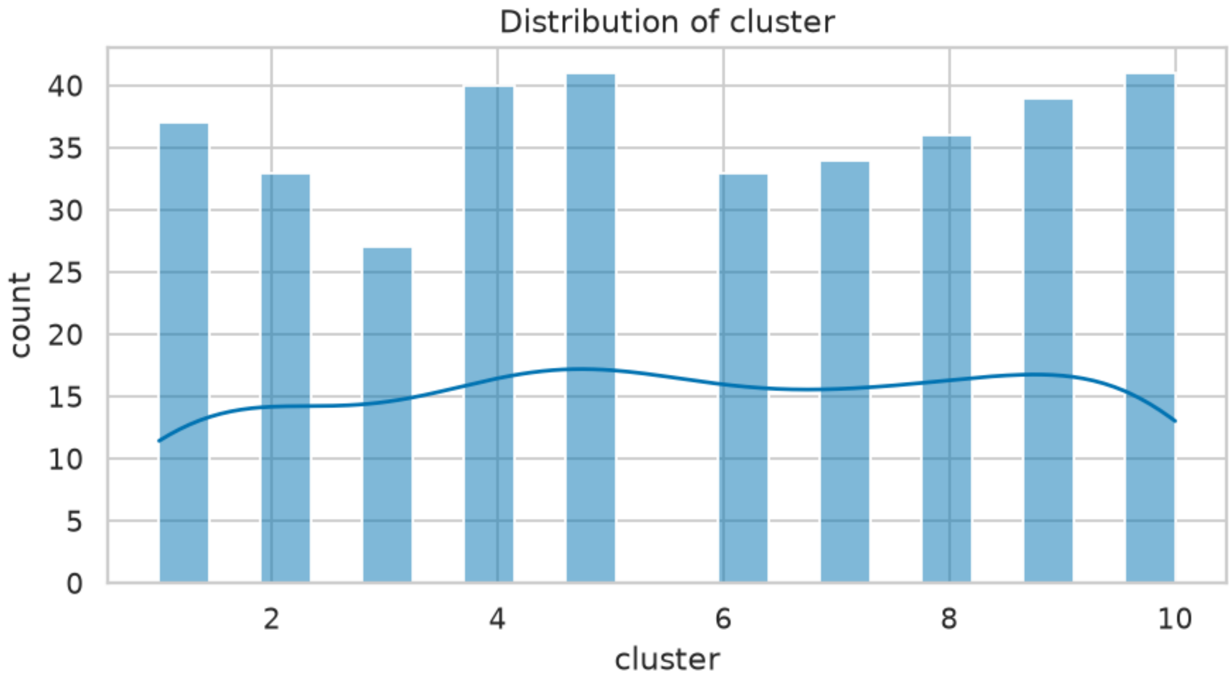

Figure 2: dist 00 cluster

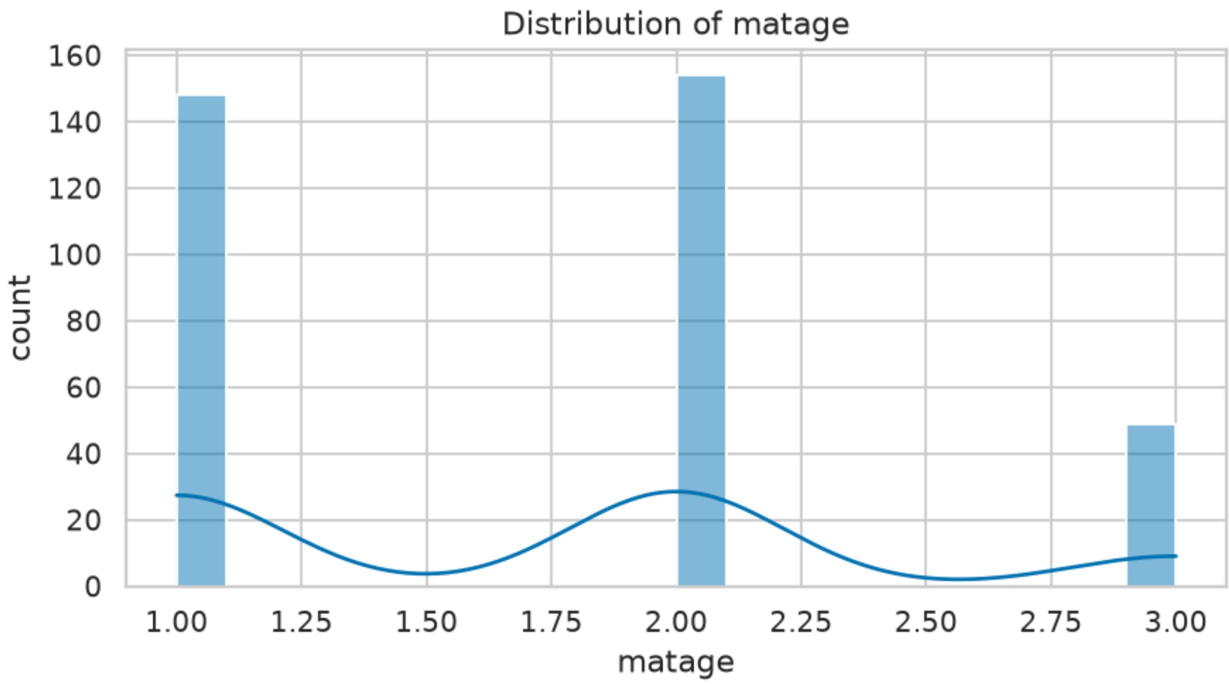

Figure 3: dist 01 matage

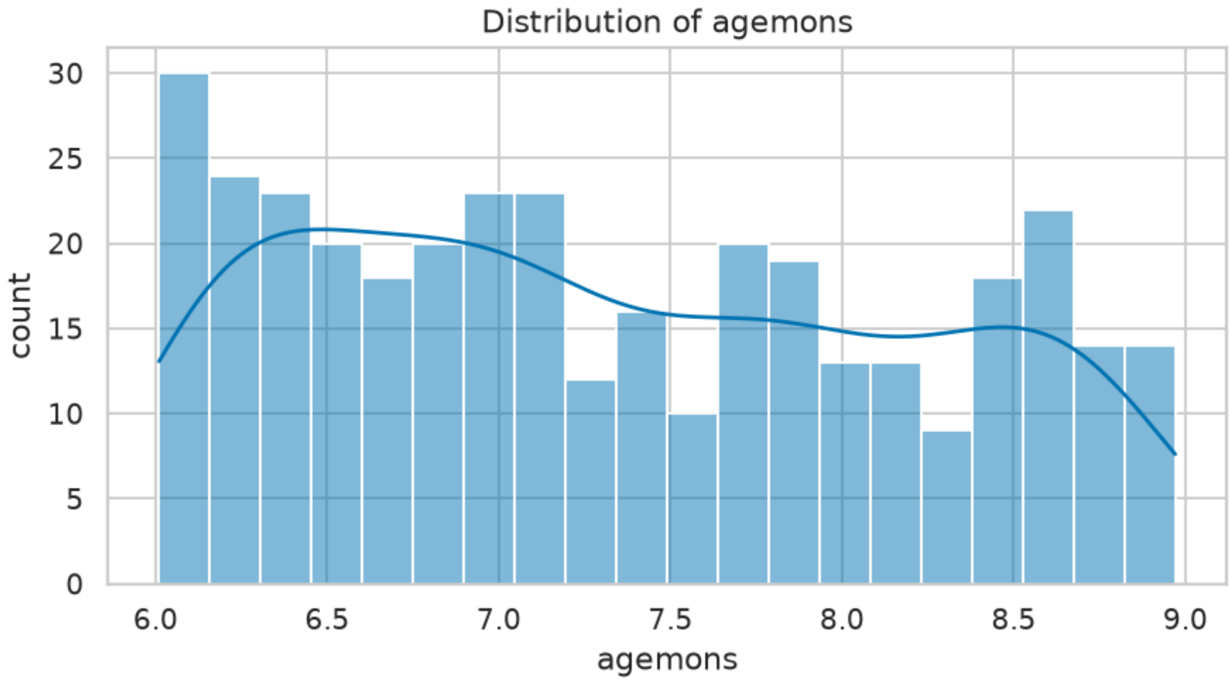

Figure 4: dist 02 agemons

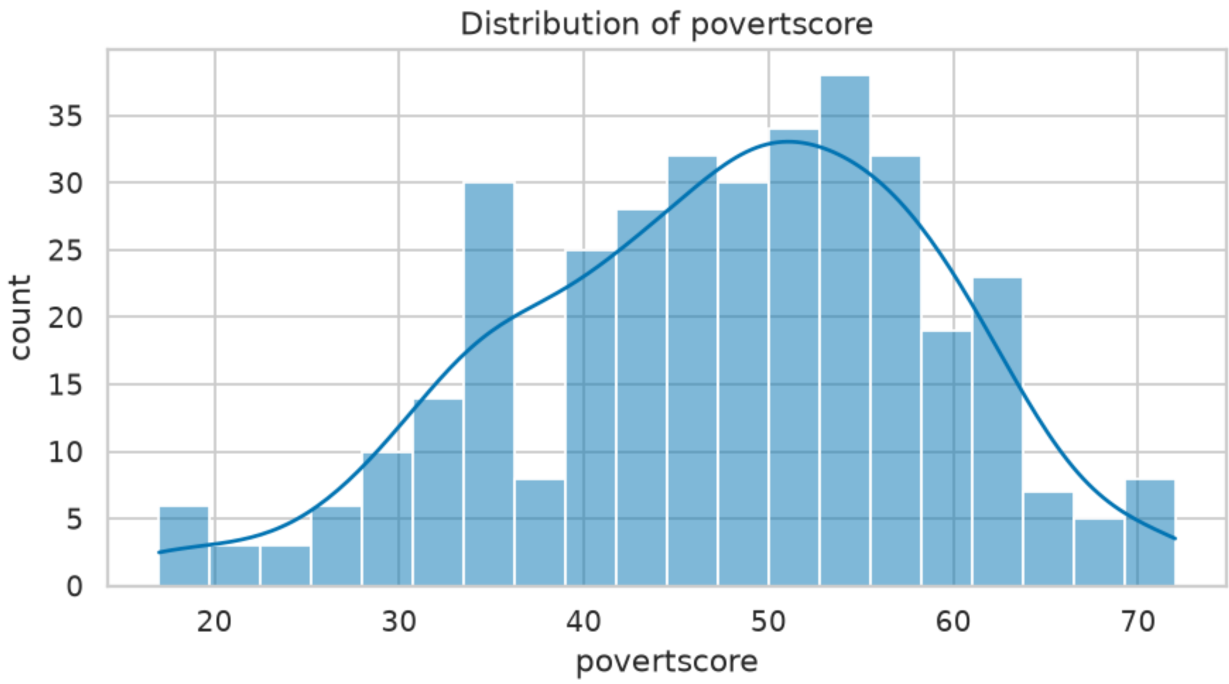

Figure 5: dist 03 povertscore

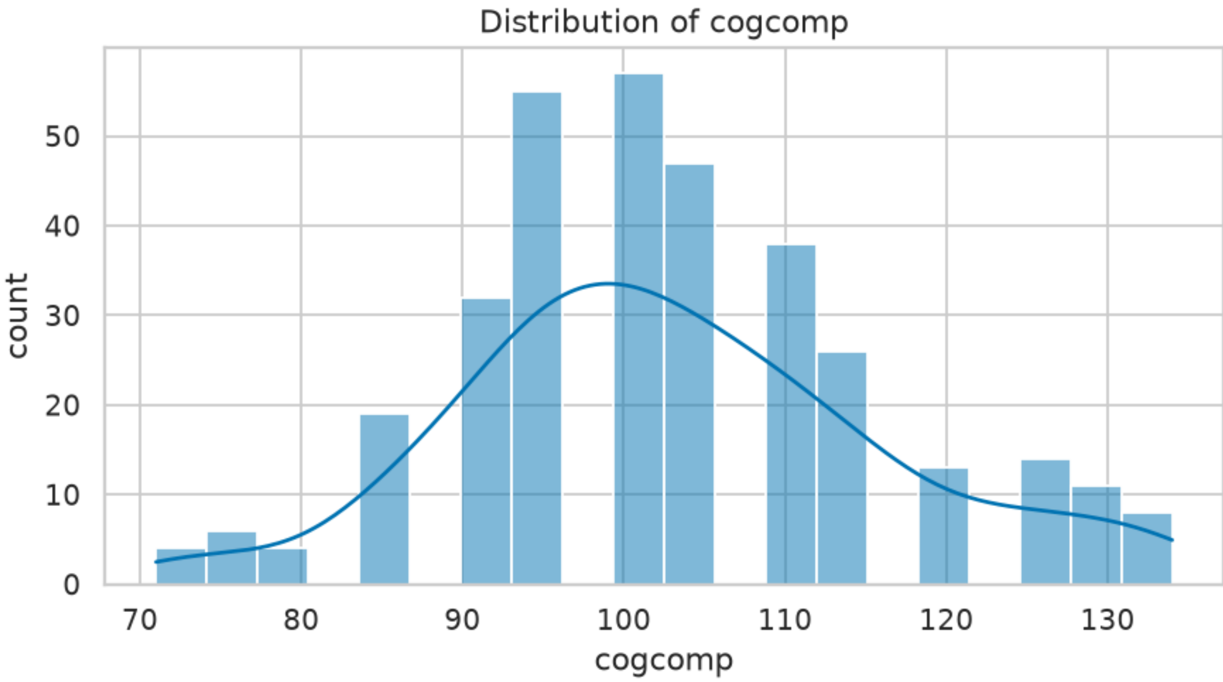

Figure 6: dist 04 cogcomp

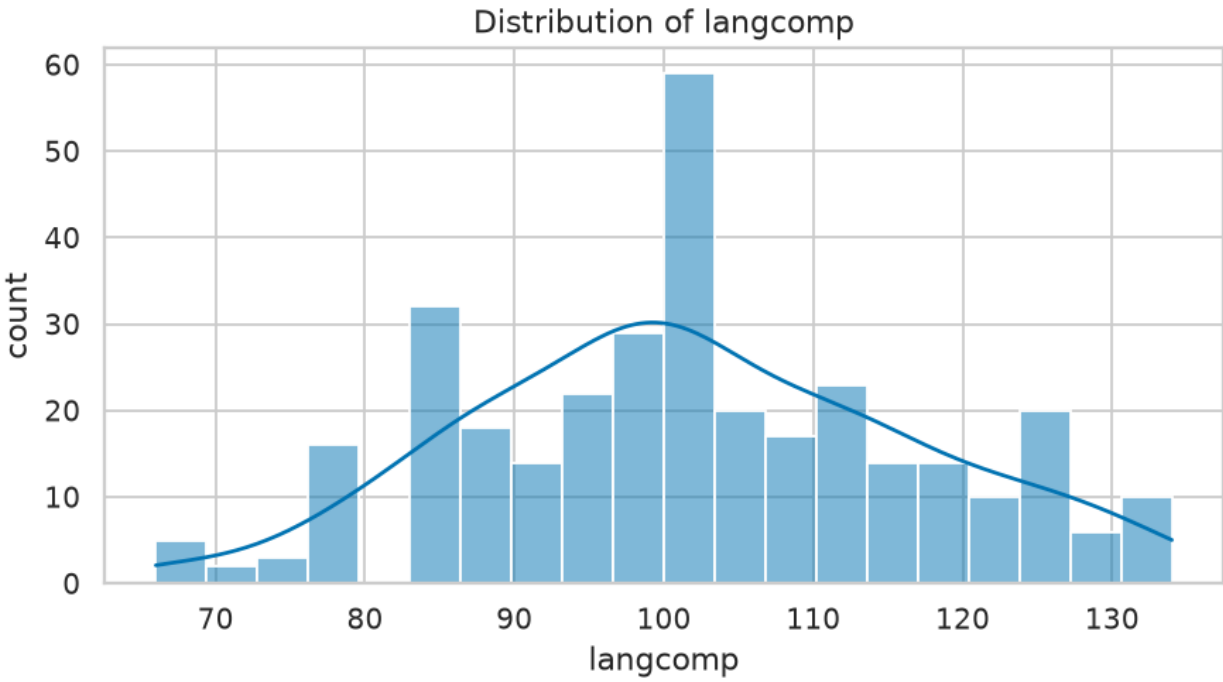

Figure 7: dist 05 langcomp

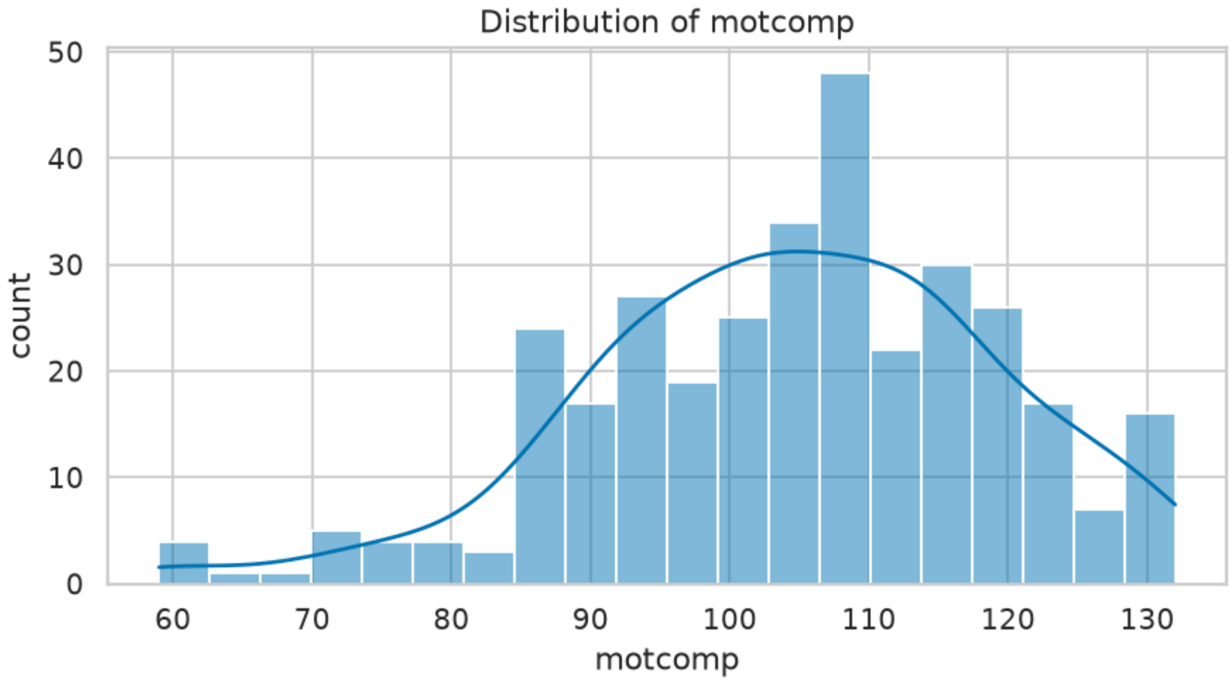

Figure 8: dist 06 motcomp

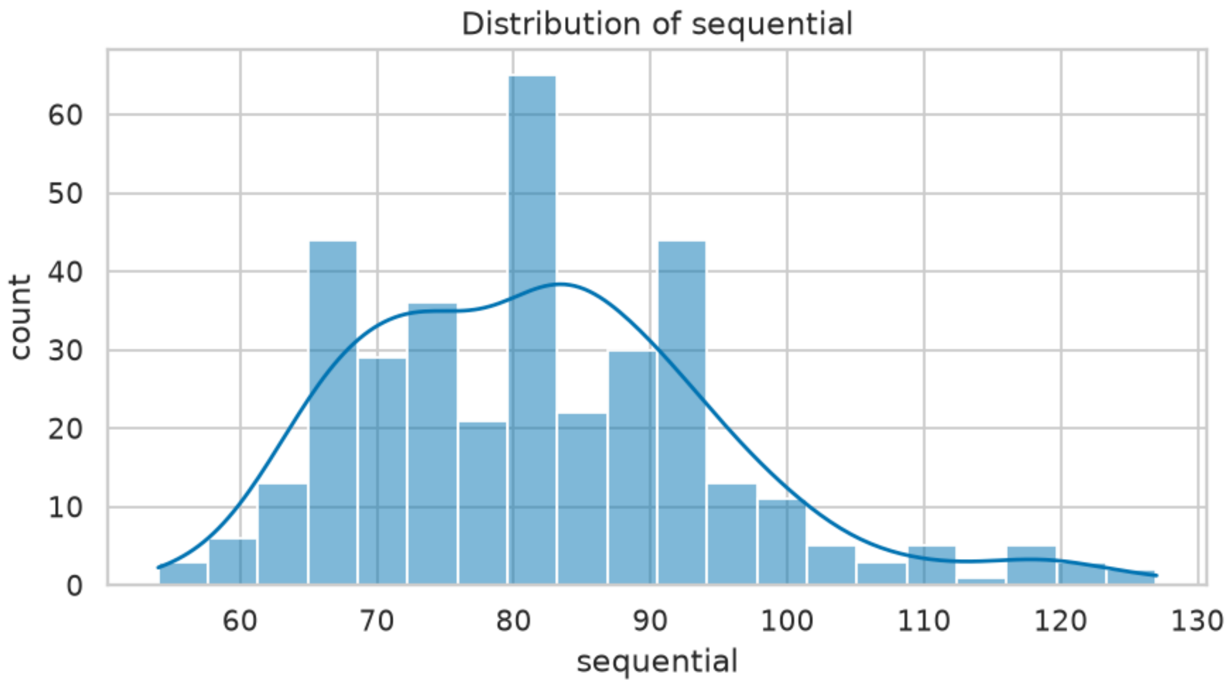

Figure 9: dist 07 sequential

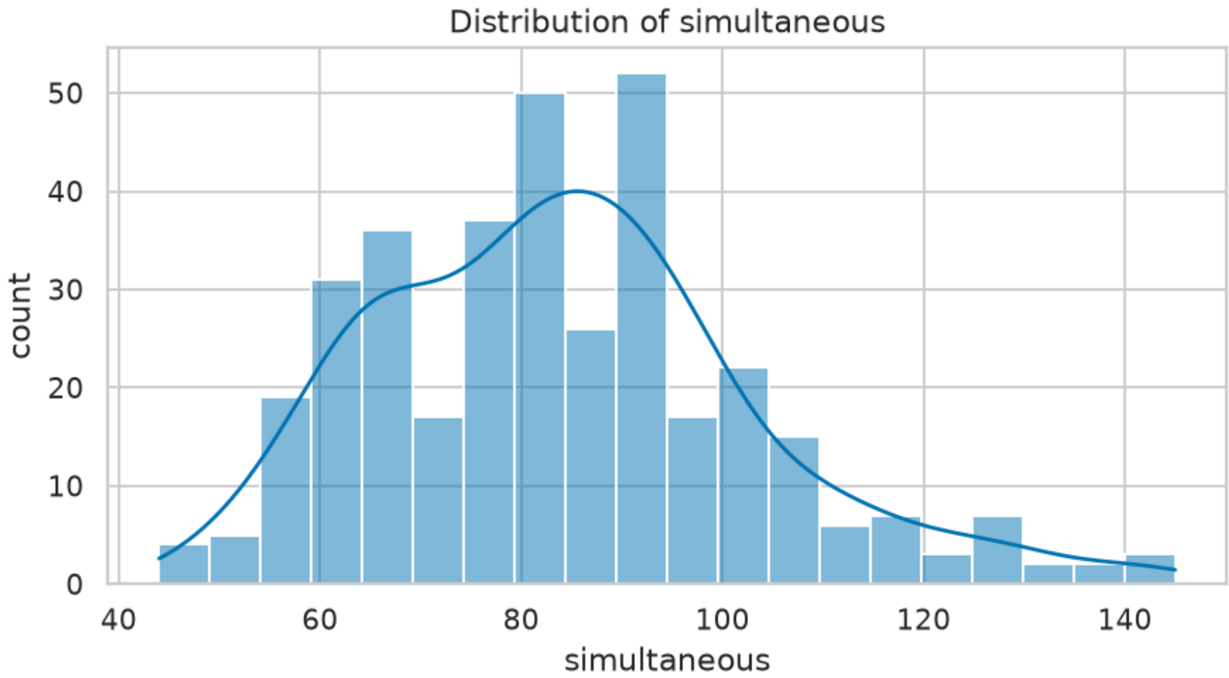

Figure 10: dist 08 simultaneous

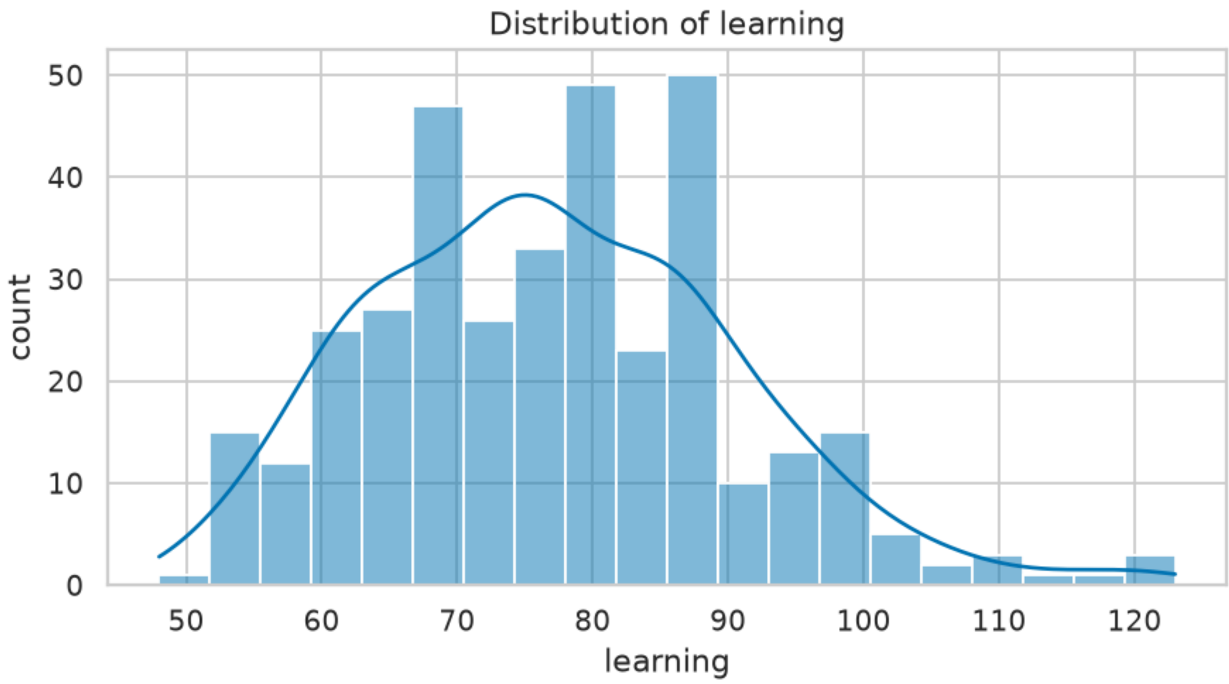

Figure 11: dist 09 learning

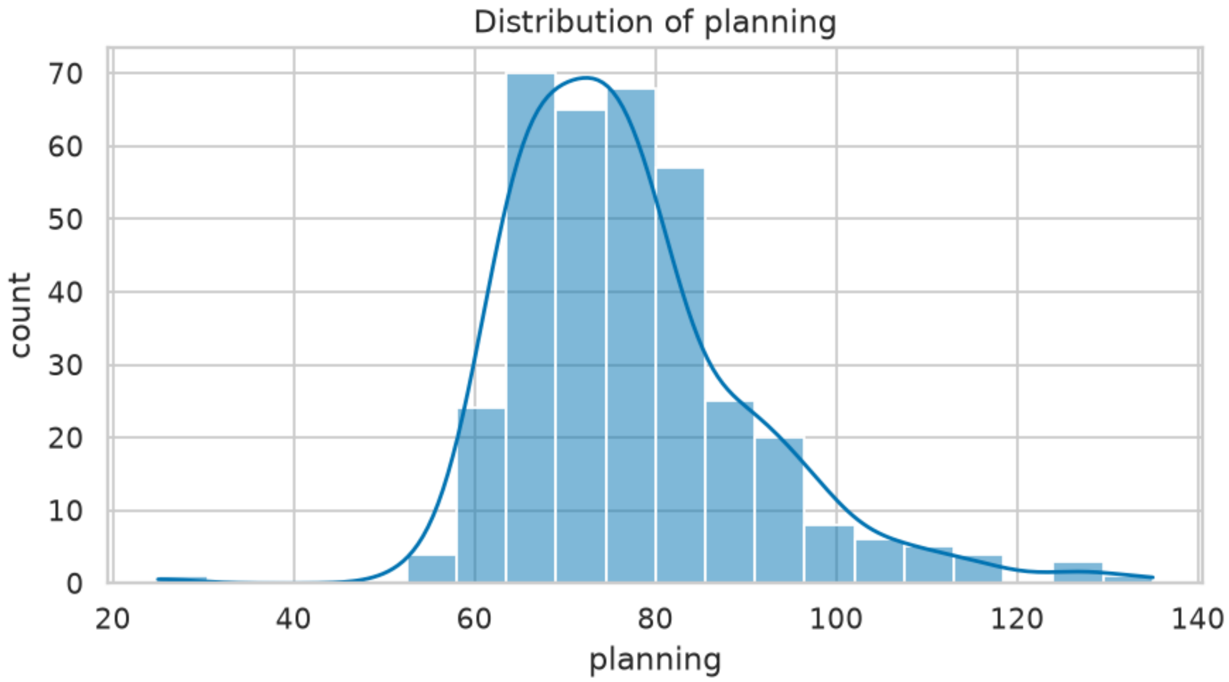

Figure 12: dist 10 planning

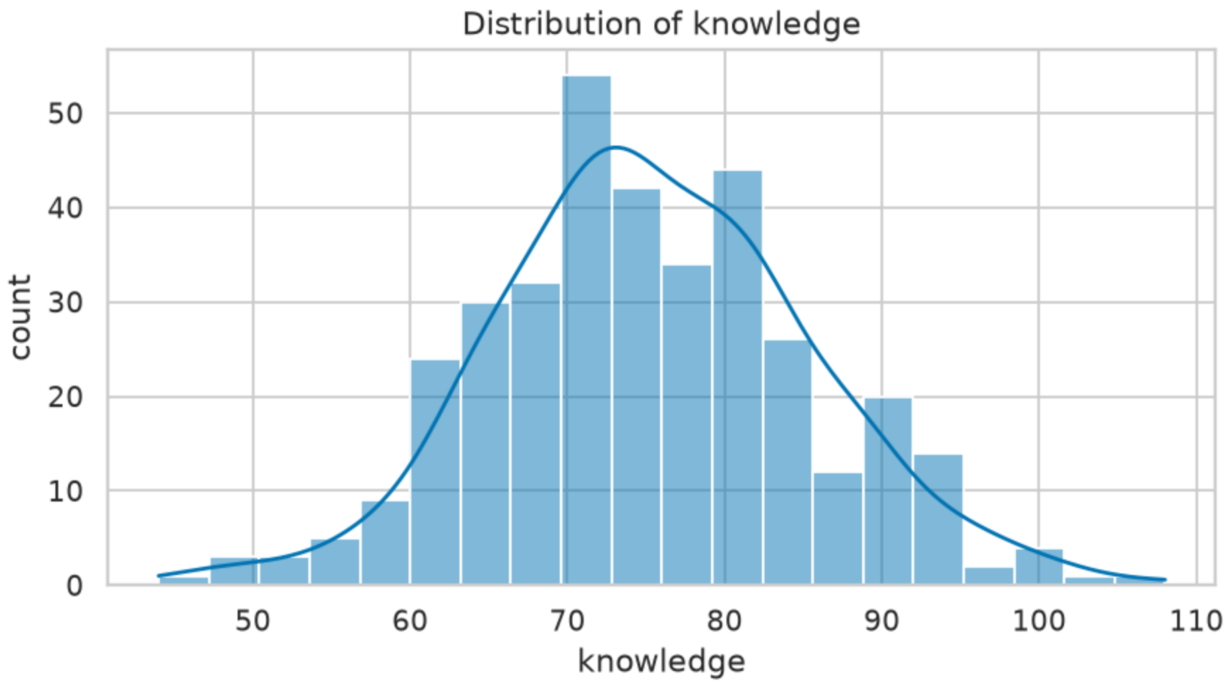

Figure 13: dist 11 knowledge

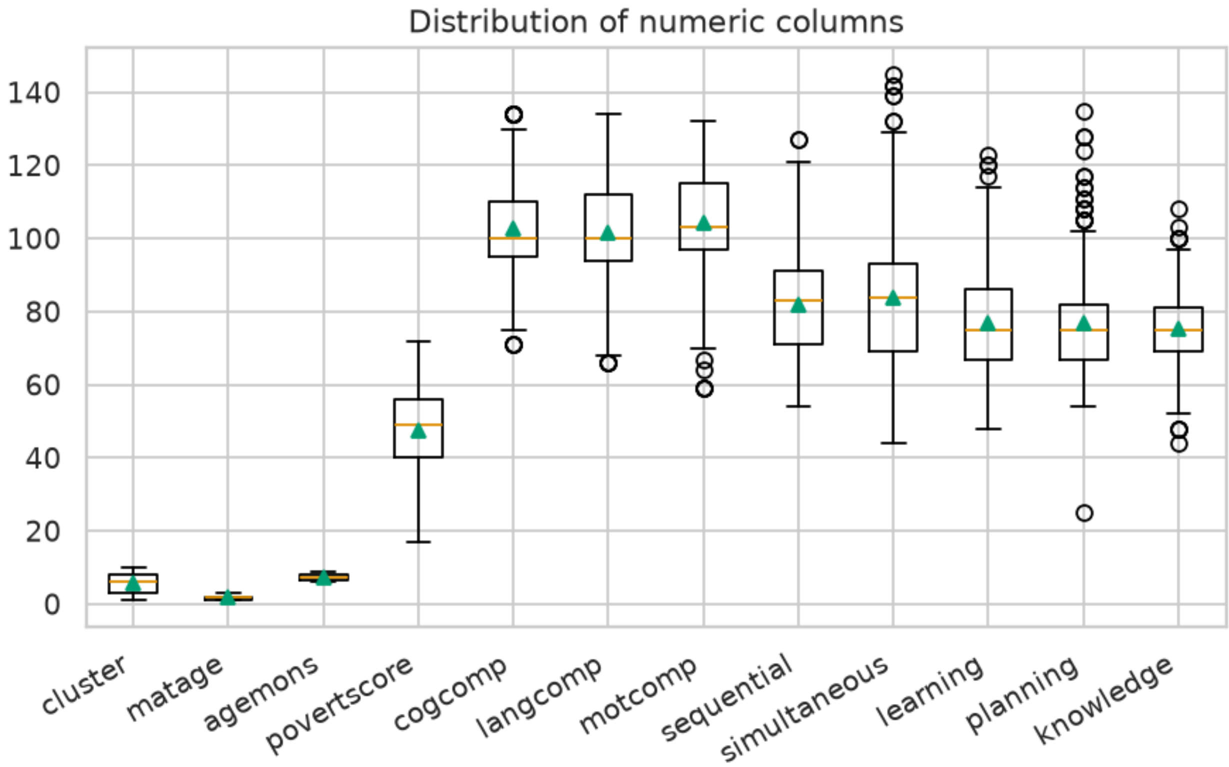

Figure 14: distributions

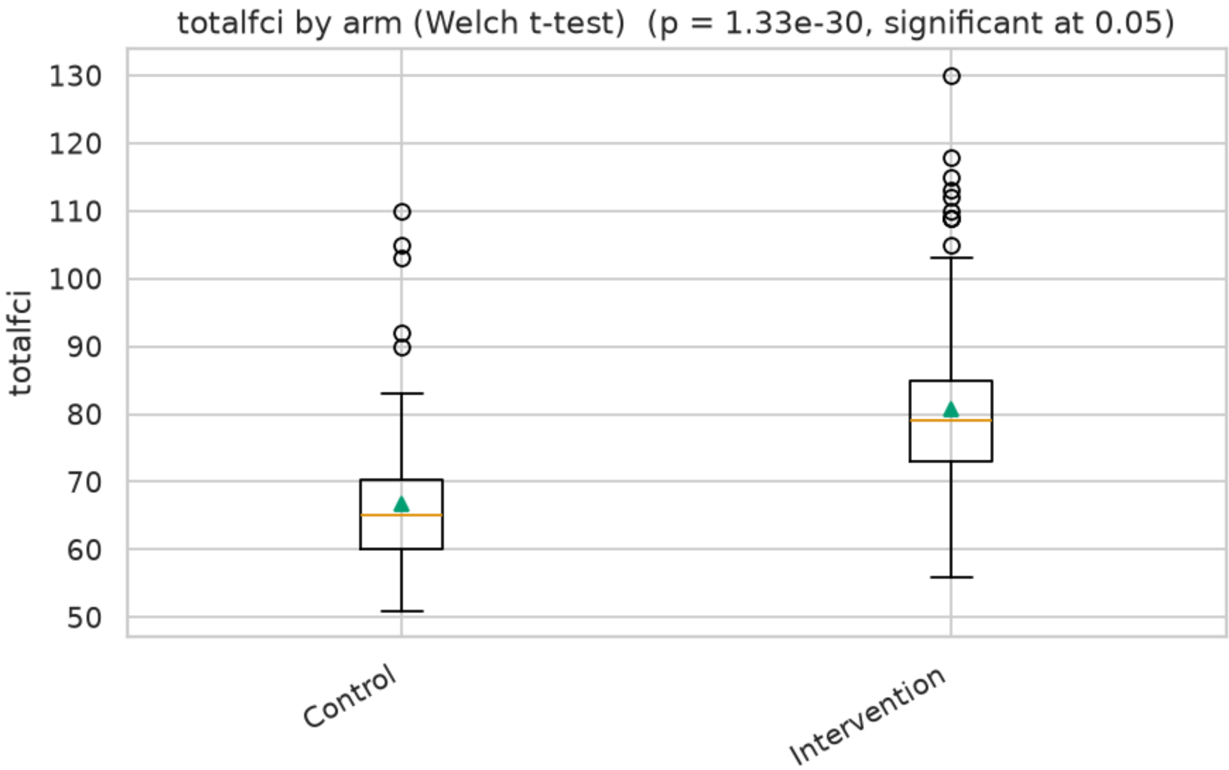

Figure 15: group comparison

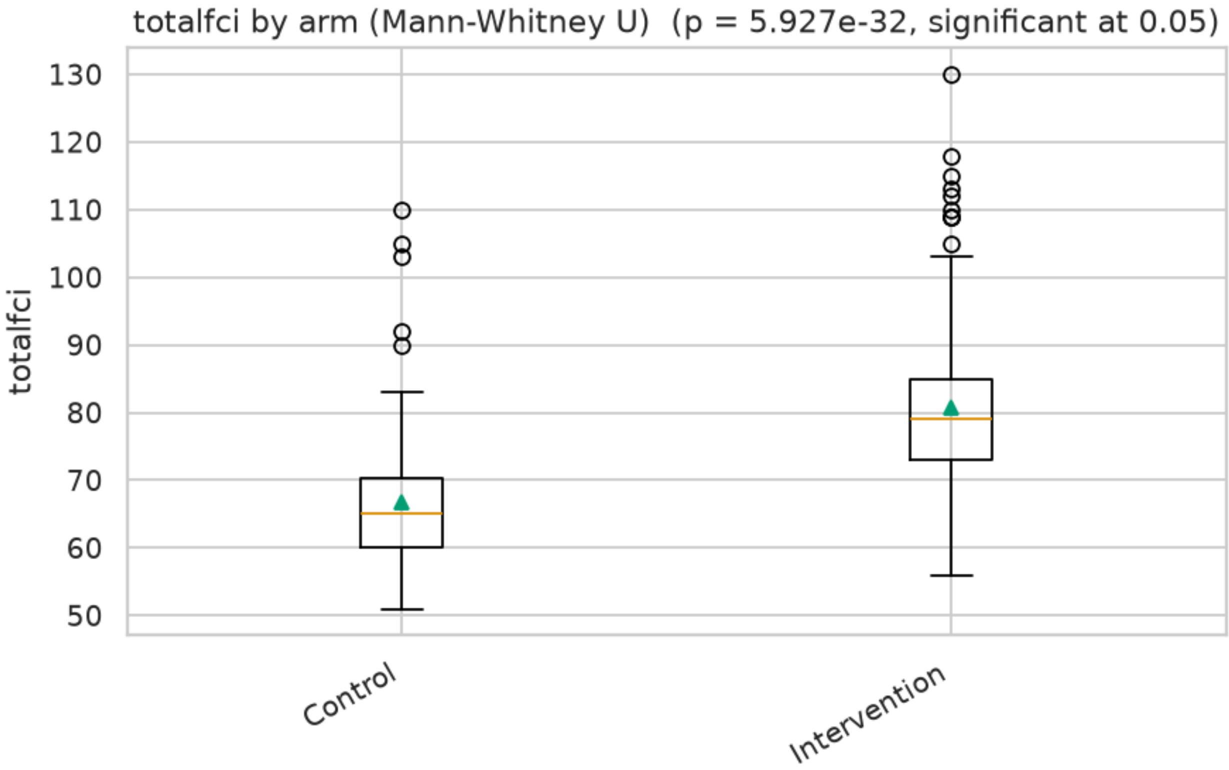

Figure 16: group comparison

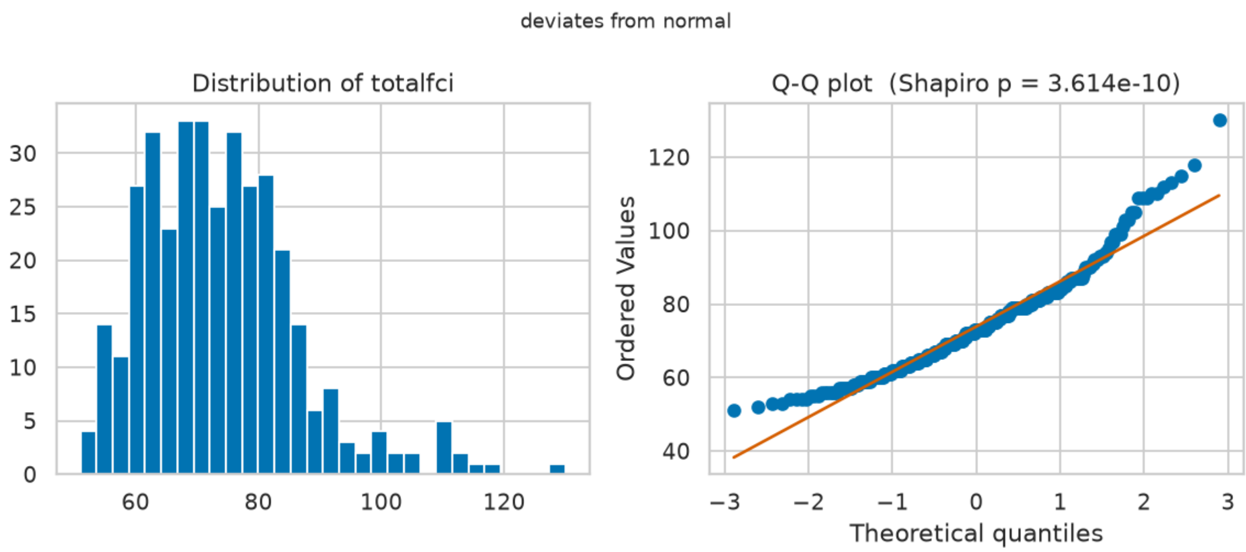

Figure 17: normality

Recorded metrics

The values below are copied directly from the run record; where the narrative cites a number, this table is authoritative.

Metric	Recorded value
input rows	361
input columns	35

plots skipped	1
distributions plotted	12
gen 1788249168235 0 / numeric columns	32
gen 1788249168235 1 / rows removed	0
gen 1788249168235 1 / duplicates removed	0
gen 1788249168235 1 / rows before	361
gen 1788249168235 1 / rows after	361
gen 1788249168235 1 / columns dropped	0
gen 1788249168235 1 / missing filled	141
gen 1788249168235 1 / fill rule	median
t-test / statistic	-12.6760
gen 1788249168235 3 / statistic	-12.6760
t-test / p value	0.0000
gen 1788249168235 3 / p value	0.0000
t-test / significant at 0.05	true
gen 1788249168235 3 / significant at 0.05	true
mann-whitney u / U	4626
gen 1788249168235 4 / U	4626
mann-whitney u / p value	0.0000
gen 1788249168235 4 / p value	0.0000
mann-whitney u / significant at 0.05	true
gen 1788249168235 4 / significant at 0.05	true
normality test / statistic	0.9463
gen 1788249168235 5 / statistic	0.9463
normality test / p value	0.0000
gen 1788249168235 5 / p value	0.0000
normality test / significant at 0.05	true
gen 1788249168235 5 / significant at 0.05	true

Limitations

The analysis is limited by the sample size and the number of simultaneous tests. Further research is needed to confirm the findings and explore the relationships between the variables in more depth. Additionally, the missingness in the dataset may affect the accuracy of the results.

> Unverified values. The following numbers appear in the narrative above but not in the run record: 280623, 30. Every value in the tables of this report is copied from the record and can be relied on; treat those in the prose as unconfirmed until checked against the run.

Produced on the AlkhemYst-Ai platform. Every figure and every value in the tables of this report comes from the recorded run; the full experiment lineage is available in the workspace.